

Customer Journey Map

Team ID: SWTID-2026-3533

Project Name: ComicCraft - AI Comic Story Creator using Gemini Models

Phase of Journey	Stage 1 Idea & Input	Stage 2 AI Story Creation	Stage 3 Comic Creation & Review
Actions What does the customer do?	• Enters a story idea or theme • Chooses genre • Gives characters and setting • Starts the creation process	• Sends the prompt to Gemini • Receives a story and dialogue • Checks whether the story matches the idea • Requests changes if needed	• Converts the story into comic scenes • Reviews panels, characters and text • Makes final edits • Saves or shares the comic
Touchpoint What part of the service do they interact with?	ComicCraft input page, prompt box, genre and character options	ComicCraft AI generation interface and Gemini model response	Comic panel editor, preview screen, edit and export/share options
Customer Thought What is the customer thinking?	"I have an idea. I want the tool to understand my story and characters."	"Will the AI create an interesting story that follows my instructions?"	"Do the panels look good? Can I easily correct anything before sharing?"
Customer Feeling What is the customer feeling?	Curious and excited to see the idea become a story.	Interested and hopeful, with some uncertainty about the AI result.	Excited and satisfied when the comic looks complete; focused while editing.
Process Ownership Who is in the lead on this?	C. Iswarya Requirement gathering, idea flow and user input	Hasini.D Gemini integration, prompt design and AI story generation	Ramya.A Comic scene/panel creation and interface flow

Opportunities
How can we improve this stage?

- Add prompt examples
- Provide easy character templates
- Give clear input guidance

- Improve prompt suggestions
- Add regeneration/edit options
- Show useful loading and error messages

- Improve panel editing
- Add preview before export
- Make sharing/export simple

Team Work Assignment

Members are assigned according to the order provided.

S.No.	Team Member	Assigned Work
1	C.Iswarya	Project requirements, user input flow, story idea/genre/character input design, and Customer Journey Stage 1.
2	Hasini.D	Gemini Models integration, prompt design, AI story generation, response handling, and Customer Journey Stage 2.
3	Ramya.A	Comic scene and panel creation, comic interface flow, character/panel presentation, and Customer Journey Stage 3.
4	Nandhini	Testing, error checking, user experience review, identifying improvements, and supporting integration.

5	Nivetha.K	Documentation, final project report, presentation preparation, final review, and coordination of project deliverables.
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Data Flow Diagram (DFD) – Project

Team ID: SWTID-2026-3533

Project Name: ComicCraft - AI Comic Story Creator using Gemini Models

Date	15 March 2026
Team ID	SWTID-2026-3533
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	2 Marks

Data Flow Diagram

The DFD shows how data moves through the ComicCraft system between the user, AI story generation process, comic creation process, and data storage. It follows the symbol and terminology guidance in the provided reference document. [filecite turn0file0 L8-L28](#)

USER	→	1. Enter Story Prompt	→	AI Story Generator
		↓		↓
		2. Generate Story	→	Gemini Models
		↓		↓
		3. Create Comic Panels	←	Generated Story
		↓		↓
		4. Save / Retrieve Comic	→	Comic Data Store
		↓		↓
USER	←	5. Display Final Comic	←	ComicCraft System

DFD Symbol Legend

Symbol / Name	Description
Oval / Rounded shape – External Entity	A person, organization, or system outside the project that sends or receives data, such as the User.
Rectangle with numbered header – Process	An activity that transforms incoming data into outgoing data, such as entering a prompt, generating a story, or creating comic panels.
Rectangle (solid fill, no number) – Data Store	A place where data is held for later use, such as saved comic data.
Labeled arrow – Data Flow	The movement of data between entities, processes, and data stores, labeled with the data being transferred.

Team Member Work Assignment

S.No.	Team Member	Assigned Work
1	C.Iswarya	Data Flow Diagram – external entity and user input flow
2	Hasini.D	AI story generation process and Gemini Models data flow
3	Ramya.A	Comic panel creation process and generated-story flow
4	Nandhini	Comic data store and save/retrieve data flow
5	Nivetha.K	Final comic display/output flow and DFD documentation

Reference basis: The uploaded template identifies the required DFD elements as external entities, processes, data stores, and labeled data flows. ■filecite■turn0file0■L10-L28■

SOLUTION REQUIREMENTS

Date: 05 October 2026 **Team ID:** SWTID-2026-3533

Project Name: ComicCraft - AI Comic Story Creator using Gemini Models

Team Members

S.No	Name	Assigned Work
1	C.Iswarya	Requirement analysis, user authentication and project documentation
2	Hasini.D	User interface design, comic creation workflow and story input module
3	Ramya.A	Gemini model integration, AI story generation and prompt handling
4	Nandhini	Comic panel generation, character/story consistency and output presentation
5	Nivetha.K	Testing, quality checking, final integration and project report preparation

Define Functional and Non-Functional Requirements

Functional Requirements define what ComicCraft should do, including comic story creation, AI-assisted generation, user interaction and output generation. Non-Functional Requirements describe how well the system should perform in terms of speed, scalability, security, reliability and usability.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users should be able to securely sign in or verify their identity before using personalized ComicCraft features.	High
2	Authorization levels	The system should provide suitable access levels for users and project/team administration, with permissions based on the assigned role.	Medium
3	External interfaces	ComicCraft should connect with Gemini models/APIs to process prompts and generate AI-based comic story content.	High
4	Transactions processing	The system should process user prompts, story inputs, AI responses and comic-generation requests in a clear and controlled workflow.	High
5	Reporting	The system should provide generated comic/story output and, where required, project or generation details for review and presentation.	Medium
6	Business rules, etc.	The system should accept a user's story idea, generate relevant content using Gemini models, organize it into comic panels, and maintain logical story flow.	High
7	Compliance to laws or regulations	The system should handle user data responsibly and avoid generating or storing unnecessary personal information; generated content should follow applicable usage and privacy requirements.	Medium
8	Other	Users should be able to enter a theme or prompt, generate a comic story, review the result and regenerate or refine content when needed.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	ComicCraft should respond to user actions quickly and process normal story-generation requests without unnecessary delay.	Main pages should load within about 2–3 seconds; generation status should be clearly shown to the user.
2	Scalability	The system should support increasing numbers of users, prompts and generated comic projects without major performance degradation.	Should support multiple users and projects with stable performance under expected academic-project usage.
3	Security & Data Privacy	User inputs, project information and generated content should be protected from unauthorized access. API credentials should not be exposed in the client interface.	Use secure authentication where applicable and protect API keys and stored user/project data.
4	Reliability & Availability	The system should handle API or generation failures gracefully and allow the user to retry without losing the current task unnecessarily.	Successful handling of normal errors with clear messages; target availability of 99% during intended use.
5	Usability & Accessibility	The interface should be simple, readable and easy for students/users to understand, with clear prompts, buttons and comic output.	A new user should be able to create a basic comic with minimal guidance.
6	Other	The application should be maintainable and portable so that project components can be updated or reused easily.	Modular code, clear documentation and compatibility with the intended development environment.

Project Team Work Allocation

The work is assigned according to the order of names provided by the team. Each member has a primary responsibility, while all members can collaborate during integration, testing and final presentation.

TECHNOLOGY STACK

Date: 05 October 2026 **Team ID:** SWTID-2026-3533

Project Name: ComicCraft - AI Comic Story Creator using Gemini Models

Define Your Technology Stack

The following technology stack is selected for ComicCraft to support an interactive web application, AI-based comic story creation, data management, deployment and team collaboration.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, JavaScript	Provides a simple, responsive and user-friendly interface for entering story prompts, displaying generated stories and presenting comic panels.
2	Backend / Server-Side	Python with Flask	Provides a lightweight backend for handling user requests, connecting the frontend with Gemini services and managing the comic-generation workflow.
3	Database / Data Storage	Firebase Firestore	Stores project information, user-related data and generated content using a flexible cloud database suitable for an application with changing content.
4	Cloud / Hosting / Deployment	Firebase Hosting	Provides convenient web hosting and deployment for the ComicCraft application with easy integration with Firebase services.
5	Version Control & CI/CD	GitHub	Supports team collaboration, source-code version control, change tracking and organized development of the project.
6	Third-Party APIs / Other Tools	Google Gemini API	Enables AI-powered story generation, prompt processing, character/story development and creative comic content generation.

Team Work Allocation

Work is assigned according to the order of names provided by the team. Members can collaborate during integration and final submission.

S.No	Team Member	Assigned Work
1	C.Iswarya	Frontend technology setup, page structure and user interface implementation.
2	Hasini.D	Backend development, server-side request handling and Gemini API connection support.
3	Ramya.A	Database setup, data storage structure and project-content management.
4	Nandhini	Deployment setup, hosting configuration and integration support.
5	Nivetha.K	GitHub repository management, testing, documentation and final technology-stack verification.